and only one overall phase to ignore. The result is that the most general state for a two-spin system has six real parameters. Evidently, the space of states is richer than just those product states that can be prepared independently by Bob and Alice. Something new is going on. The new thing is called *entanglement*.

Entanglement is not an all-or-nothing proposition. Some states are more entangled than others. Here is an example of a maximally entangled state—a state that's as entangled as it can be. It is called the *singlet* state, and it can be written as

$$|sing\rangle = \frac{1}{\sqrt{2}}\big(|ud\rangle - |du\rangle\big).$$

The singlet state cannot be written as a product state. The same is true for the triplet states,

$$\frac{1}{\sqrt{2}}\big(|ud\rangle + |du\rangle\big)$$

$$\frac{1}{\sqrt{2}}\big(|uu\rangle + |dd\rangle\big)$$

$$\frac{1}{\sqrt{2}}\big(|uu\rangle - |dd\rangle\big),$$

which are also maximally entangled. The reason for calling them singlet and triplet will be explained later.

Exercise 6.3: Prove that the state |*sing*> cannot be written as a product state.